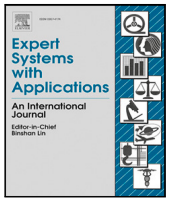
Expert Systems With Applications 256 (2024) 125006



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa





An LDoS attack detection method based on FSWT time–frequency distribution☆

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ARTICLE INFO

Keywords:
LDoS attack
Time–frequency domain
FSWT
Time–frequency distribution
Decision tree

ABSTRACT

The Low-rate Denial-of-Service (LDoS) attack is a stealthy and periodic attack, which belongs to the category of DoS attacks. The LDoS attack maliciously preempts and consumes target resources, causing the targeted network performance to decline and affecting the service quality. The LDoS attack is highly destructive and difficult to detect and defend against due to its abnormal attack behavior. In this paper, we discuss the network traffic behavior in the time domain, frequency domain, and time–frequency domain, and we find that the time–frequency domain contains more detailed information than the time domain and frequency domain alone. Considering the limitations of the existing time–frequency domain transformation methods, an LDoS attack detection method based on Frequency Slice Wavelet Transformation (FSWT) is presented in this paper. The entropy, ratio of energy, contrast, and correlation are extracted from the time–frequency distribution to depict the network traffic, and a decision tree is trained to detect the LDoS attack in this paper. According to the experimental results on NS2, testbed, and the comparative experiment, we conclude that the method presented in this paper performs well.

1. Introduction

The healthy development of Internet and security is threated by network attacks, and Denial-of-Service (DoS) attack is among one of the most widely distributed and threatening network attacks. Nowadays, DoS and Distributed Denial-of-Service (DDoS) attacks are widely reported on the Internet. Since the 21st century, a new DoS attack has appeared on the Internet, and the attack behavior is different from the traditional DoS/DDoS because of its periodic, stealthy, and low-rate characteristics (Jian, Mei, & Junyao, 2015). For the first time it was proposed, it was called the shrew attack (Kuzmanovic & Knightly, 2003), and later it was named the Reduction of Quality (RoQ) attack (Guirguis, Bestavros, & Matta, 2004), or the Pulsing Denial-of-Service (PDoS) attack (Luo, Chang, et al., 2005). For convenience, this type of attack is referred to as the Low-rate Denial-of-Service (LDoS) attack in this paper.

LDoS attack is a novel denial-of-service method. Unlike the DoS/DDoS attacks, this kind of attack does not require a high-rate attack flow to flood or exhaust the bandwidth and resources of the targeted links or hosts. Instead, it uses vulnerabilities within network protocols or adaptive mechanisms of application services, periodically sending attack traffic to reduce the TCP service quality and throughput. LDoS is aimed at the TCP congestion control mechanism, e.g., the retransmission time out (RTO), as well as the additive increase and multiplicative decrease (AIMD). As a result, the TCP transmission performance in the network is deteriorated, and the network service quality is affected significantly.

Due to the low-rate attack pattern, LDoS attacks can be easily hidden in background traffic, making it challenging to detect. Currently, most of the time-domain-based detection methods for LDoS attacks only utilize a network multi-feature fusion (Yue, Wang, Liu, & Wu, 2020), and even do not incorporate the frequency-domain analysis (Chen, Hwang, & Kwok, 2005). On one hand, the only time-domain or frequency-domain analysis cannot completely characterize LDoS attack features, reducing the accuracy of the LDoS detection. On the other hand, the detection approaches utilizing the network multi-feature fusion rely on machine learning models and thus exhibit poor interpretability (Tang, Wang, Li, Vijayakumar and Kumar, 2023). Although these approaches have acceptable detection accuracy, they

☆ This work is supported by YueLuShan Center Industrial Innovation Project (#2023YCII0115), as well as Science and Technology Key Projects of Changsha City (#kq2208038).

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https://doi.org/10.1016/j.eswa.2024.125006
Received 12 January 2024; Received in revised form 1 July 2024; Accepted 4 August 2024
Available online 6 August 2024
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only analyze the abnormal network status caused by LDoS attacks from a long-term perspective and cannot grasp the essence of LDoS attack flows. In general, we encounter the following issues and difficulties while attempting to identify LDoS attacks:

- How to robustly analyze the time–frequency characteristics to model this concealed attack?
- How to grasp the essence of LDoS attacks and extract effective features for attack detection?
- How to balance the robustness, accuracy and system overhead of the proposed detection approach?

To address these problems, we propose a detection method based on Frequency Slice Wavelet Transformation (FSWT) to identify LDoS attacks. Based on FSWT, we transform the original time domain network traffic into the time–frequency domain format, i.e., a time–frequency distribution. Subsequently, we analyze the characteristics from the distribution and leverage a decision tree model to differentiate between LDoS attack and legitimate traffic.

We conduct detection experiments in different environments, including NS2, a physical testbed, and SDN, achieving detection accuracies of approximately 96%, 99%, and 95%, respectively. The results prove that FSWT is feasible and effective for detecting LDoS attacks. The features extracted by FSWT are comprehensive and effective, and can be used to distinguish normal and LDoS attack status. The main contributions of this paper are as follows:

- We use FSWT to comprehensively analyze the abnormal status of the network caused by LDoS attacks.
- We integrate the network time domain and frequency domain features, and extract relevant information based on FSWT to detect LDoS attacks.
- We implement a low-overhead and highly accurate LDoS attack detection method according to the characteristics of the network.

This paper is organized as follows: Section 2 introduces the threat model of LDoS attacks. Section 3 discusses the general research process of the LDoS attack and the existing detection methods. Besides, an LDoS attack detection method based on FSWT time–frequency distribution is described in Section 4. The method is validated through different networks established on different platforms in Section 5. In Section 6, we have discussed some broader network security research and new detection methods. We summary the current work and conclude this paper in Section 7.

2. Background

The LDoS attack is one new type of DoS attack. There are two main subjects involved in this attack, which are the active initiator and the passive defender, corresponding to the attackers and victims respectively. There may be one or more attackers, and the victim may be a server, a link, or a network. As shown in Fig. 1, the attack pulses are sent periodically by LDoS attackers through three parameters (T, L, R), where T represents the interval of each pulse, L is the duration of each attack pulse, and R denotes the rate of each pulse. To achieve a stealthy attack, the attack duration L is set shorter than the attack interval T ($L < T$), and the LDoS attacker stays silent in the interval duration T-L. By the adoption of periodic launching, it ensures the low-rate characteristics of the LDoS attack. Thus, in a network with the LDoS attack, the average malicious traffic is $R * \frac{L}{T}$. When the attack duration L is approximately equal to the attack interval ($L \approx T$), the attack is the well-known DoS/DDoS attack.

The ideal LDoS attack is a balance of the three parameters. On one hand, it cracks down the victim network and degrades the service quality seriously. On the other hand, it is too concealed to be detected and filtered by the traditional DoS/DDoS detection mechanism.

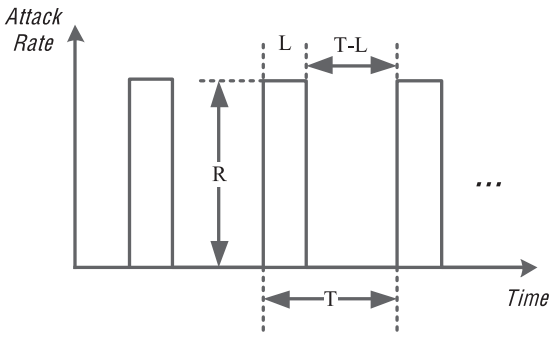


Fig. 1. The LDoS attack parameter model.

3. Related work

In recent years, many researchers have attended research on the LDoS attack. Their research areas include, but are not limited to, traditional Internet, SDN (Tang, Yan, Zhang, Chen and Qin, 2022; Tang, Zhang, Yan, Chen and Qin, 2022; Yungaicela-Naula, Vargas-Rosales, Pérez-Díaz, & Carrera, 2022), WSN (Chen, Liu and Zhongchuan, 2019; Chen, Meng, Shan, Fu and Bhargava, 2019; Liu, Sun, Zhang, & Li, 2023), Cloud platform (Agrawal & Tapaswi, 2018), IoT (Okada, Miyamoto, Sekiya, & Nakamura, 2021), etc.

3.1. LDoS attack research stage

The research process consists of three stages: analyzing attack behavior, developing detection technology, and establishing defense strategy. Generally, researchers research one, two, or all of the three stages.

Attack behavior. The attack behavior is generally divided into single-source LDoS attack and multi-source LDoS attack. In particular, within the latter, multiple LDoS attackers initiate attack behaviors synchronously (Wu, Pan, Yue and Liu, 2019; Yue, Wu, & Lei, 2014) or asynchronously (Wu, Pan et al., 2019). Currently, many researchers model LDoS attack behaviors (Luo et al., 2014; Macia-Fernandez, Diaz-Verdejo, & Garcia-Teodoro, 2009; Tang, Luo, Hui, & Chang, 2014) to better analyze the attack characteristics.

Detection technology. The detection technology can be classified into anomaly detection and feature detection. Detection standards are established through normal behaviors to detect abnormal behaviors in anomaly detection. Its detection process is generally to train normal data to obtain a detection threshold, and then compare the abnormal data with the threshold to obtain detection results (David & Thomas, 2019; Thangavel & Kannan, 2022; Wu, Pan et al., 2019). Feature detection acquires a feature set of normal behavior and abnormal behavior in advance, and then the data to be tested is analyzed to observe which behavior it matches better. The most commonly used method for feature detection is algorithms related to classification, such as machine learning algorithms (Tang et al., 2020). An AI-based intrusion detection and prevention system (Chang, Boddu, Xu, & Doan, 2023) can integrate multiple learning models to generate attack alerts for operators, thereby blocking unauthorized users.

Defense strategy. It is mainly achieved by filtering the attack flows or developing corresponding network services. Filtering attack flow refers to filtering suspicious traffic directly based on IP and port. GASF-IPP (Tang, Wang, Liu, Jin and Zhang, 2023) is an LDoS attack detection and defense system in SDN. After detecting an attack, it establishes a list of suspicious IPs and ports based on IP ratio and port variation, and installs this list as a blacklist in the flow table of switches. Subsequent packets matching this flow table are directly discarded. Additionally, developing network services to defend against is relatively more sophisticated. They are often achieved by revising some existing network protocols or router queue management algorithms (Xie, Xu, Cao, & Li, 2019), which is significantly hard to follow.

3.2. Feature extraction

The features obtained from the LDoS attack analysis can be used to create the detection methods. Considering the different way of feature extraction, they fall into one of the three following categories.

Time domain. To achieve the goal of the LDoS attack detection, researchers can make use of multiple intrinsic time-domain features (TCP energy, TCP proportion, UDP energy, UDP proportion, etc.). By fusing these time-domain features into a feature map, these time-domain features are directly used as the input of the improved Adaboost algorithm to classify normal traffic status and the abnormal network status under LDoS attacks (Tang et al., 2020). Focused on the periodicity of pulsed attack flow, Wu, Pan et al. (2019) constructs a local sequence and judges it by a similarity score compared with background traffic. ERT-EDR (Liu et al., 2024) is deployed on the SDN controller, querying switches regularly to obtain traffic time series. It identifies LDoS attacks by measuring both the coefficient of variation and Spearman's rank correlation of the traffic.

In general, the analysis based on the time domain is more intuitive because the performance of network traffic is direct and easy to understand on the time domain. Besides, it is fast and convenient to extract statistical characteristics (such as the max value, the mean value, the variance, and the proportion of traffic, etc.) directly based on the time domain. However, the LDoS attack flow is a periodic signal and its periodicity is difficult to be reflected in the time domain. Different network behaviors may have the same characteristics in time domain statistical features, causing more false results.

Frequency domain. Fast Hartley Transform has less computation requirement than traditional Fourier transform algorithms, and it can be used to analyze the power spectral distribution of the traffic. When combined with the assessment of the attacker's actions, such as delivering periodic pulses, Agrawal and Tapaswi (2020) creates a precise and lightweight method for detecting low-rate DDoS attacks. Additionally, as mentioned in Chen, Yeo, Lee, and Lau (2018), power spectrum analysis is coupled with information entropy derived from Fourier Transform (FT) and Wavelet Transform (WT) to counter LDoS attacks. According to Wu, Wang, Yan, and Yue (2017), a comb filter can block attack traffic and release legitimate traffic based on variances in energy distribution at specific points. The specific points are referred by frequency domain researching method expected to be n/RTT. Furthermore, in a real-world cloud architecture, Agrawal and Tapaswi (2018) demonstrates the effectiveness of frequency-domain approaches in defending against both internal and external multi-source LDoS attacks.

In general, the time domain network traffic is usually transformed into the frequency domain format, i.e., a group of frequency components. With the spectrum analysis, the statistical characteristics of these components (such as centroid frequency, mean frequency) are extracted to better analyze the changes in the network behavior. Especially, the periodicity of LDoS attacks can skew the frequency components in the network traffic. So the frequency domain characteristics of the network traffic can accurately characterize the changes brought by the LDoS attacks in the spectrum. However, the disadvantage is that the frequency-domain-based transform cannot provide the timing dependency information. Since each signal is processed as a whole, two distinct signals in the time domain could consist of the same frequency components, thus the LDoS attack behavior cannot be adequately described by the frequency domain properties alone.

Time–frequency domain. The Short-Time Fourier Transform (STFT) can be used to retrieve the time–frequency domain information of network traffic. Furthermore, it is noted that the time–frequency joint distribution of TCP traffic varies depending on the condition of the network and performs differently under the situation with or without LDoS attacks. Based on the above consideration, Fu, Duan, Wang, and Li (2022) filtered LDoS attack traffic out according to the obvious reconstruction error through a neural network trained by time–frequency features of normal traffic. According to Chen, Meng et al. (2019), the study of non-stationary tiny signals resulting from the LDoS attack can be effectively conducted using the Hilbert–Huang Transform (HHT) time–frequency analysis. In this work, the researchers focus on ZigBee Wireless Sensor Network (WSN) and check each component computed by HHT. Here, the trustworthy components are picked out to evaluate the time range and suspicious network nodes related to LDoS attacks. Notably, the HHT method can be improved and embellished to be more adaptive to the specific scenarios. Accounting for the randomness of RoQ attack, Chen, Liu et al. (2019) uses Ensemble Empirical Mode Decomposition (EEMD) to overcome the potential risk of mode mixing in HHT. And assisting by correlation coefficient, it screens out the false IMF component of routing message traffic to shrink the detection error rate.

In order to properly reflect the detailed characteristics in both the time and frequency domains, many researchers currently prefer to employ the time–frequency joint analysis approach to process network traffic and extract the local particular information (Yue, Liu, Wu, & Wang, 2018). In other words, the time–frequency domain analysis can make up for the limitations of the analysis methods based on a single domain. In this paper, we proposed an FSWT time–frequency distribution based LDoS attack detection method. To accurately detect LDoS attacks, robust characteristics can be inferred from the time–frequency distribution of network data, as acquired using FSWT.

4. An FSWT time–frequency distribution based LDoS attack detection approach

There are some limitations of known time–frequency transformation, such as Short-Time Fourier Transform (STFT) and Wavelet Transform (WT). STFT is a commonly used time–frequency algorithm. It applies a sliding time window to the signal and performing a Fourier Transform (FT) on the signal within the time window. The time–frequency resolution of its time–frequency distribution depends on parameters such as the window function, window length, and the moving step of the sliding window. Due to the dependency on these parameters, it is challenging to predict the optimal parameters in advance in practical applications, making its operation more complicated. More importantly, once the parameter settings are determined, the window function and window length of the STFT transform are fixed, meaning that the window lacks the adaptability.

The window of the WT is adaptable in comparison to the STFT, providing high time resolution for high-frequency signals and high frequency resolution for low-frequency signals. Although WT is not constrained by a fixed time window, the effectiveness of the obtained time–frequency distribution is still limited by the choice of wavelet basis and the length of the scale sequence. Since different wavelet basis functions and scale sequence lengths significantly affect the time–frequency distribution, it increases the difficulty of its application in obtaining the time–frequency distribution of network traffic.

To this end, FSWT is introduced to counter the defects of the two mentioned time–frequency transformation approaches. FSWT is an important method in time–frequency analysis, with good time–frequency resolution and can deeply reflect the characteristics of the signal through the distribution in the two-dimensional (Yan, Miyamoto, & Jiang, 2009, 2011; Yan, Miyamoto, Jiang, & Liu, 2010).

4.1. FSWT transformation

FSWT is a novel approach for the time–frequency analysis, proposed by Z. Yan et al. in 2009 (Yan et al., 2009). Compared with WT transform, it has more properties (Yan et al., 2010). FSWT adopts the Frequency Slice Function (FSF) and a new scale parameter. The time and frequency domain signals can be modeled as a joint distribution, and the time–frequency resolution in both low and high frequency band can be taken into account. Based on combining the advantages of STFT and WT, the FSF has an asymmetric structure in both time

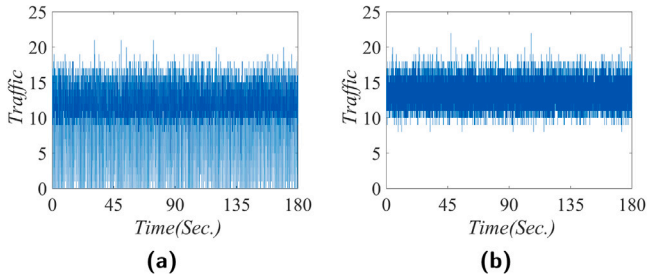


Fig. 2. Network traffic in different status. (a) is the network traffic in NS, and (b) is the network traffic in LS.

and frequency domain, so that FSWT time–frequency window center can be regarded as the frequency observation center. In addition, the FSWT time–frequency resolution can be flexibly adjusted through the scale parameter.

According to Yan et al. (2009), for a discrete sequence $X = (x_0, x_1, \dots, x_{N-1})^T$ of length N, the FSWT transformation is as follows.

$$W(,i) = \lambda F^{-1}\{F\{X\} \cdot * P_i\}, i = 0, 1, \dots, N-1 \tag{1}$$

where F stands for the Fourier transform, and F^{-1} represents the Fourier inverse transformation. The energy coefficient $\lambda \neq 0$ and $A \cdot * B$ is the product of array A and B. P_i represents the i_{th} Frequency Slicing Sequence (FSC) with a length of N.

According to the information in Yan et al. (2009), FSC is a discrete distribution of FSF, and its representation is as follows.

$$P = (0, \dots, 0, \hat{p}_{-m} \neq 0, \dots, \hat{p}_{-1}, 1, \hat{p}_1, \dots, \hat{p}_m \neq 0, 0, \dots, 0)^T \tag{2}$$

where $P_0 = (1, \hat{p}_1, \dots, \hat{p}_m, \dots, 0)^T$, $\dots$, $P_{N-1} = (0, \dots, 0, \hat{p}_{-m}, \dots, \hat{p}_{-1}, 1)^T$, the FSF is a function that has an asymmetric structure in both time and frequency domain. The most commonly used FSF are: $\hat{p}(\omega) = e^{-\omega}$, $\hat{p}(\omega) = e^{-0.5\omega^2}$, $\hat{p}(\omega) = \frac{1}{1+|\omega|^n}$.

In our approach, the network traffic time–frequency domain characteristics are expressed through FSWT. The Heisenberg uncertainty principle states that the time-domain resolution and frequency-domain resolution cannot both be optimized at the same time, and they need to be compromised through the scale parameter. By using FSWT, we transform the network traffic as the following steps and finally obtain the time–frequency domain characteristics.

(1) Preprocess the network traffic to remove the DC part in the traffic.

(2) Select the frequency resolution ratio η (default to 0.05).

(3) Choose an expected response ratio of FSWT amplitude υ (default to $\sqrt{2}/2$, 1/2, or 1/4).

(4) Select FSF and calculate the parameter κ based on the inequality, i.e., $|\hat{p}(\kappa\eta)| \leq \upsilon|\hat{p}(0)|$ and $\left|p(\frac{\mu}{\kappa\eta})\right| \leq \upsilon|p(0)|$ according to Yan et al. (2009).

(5) Perform time–frequency transformation according to Eq. (1) to obtain the time–frequency transformation energy distribution matrix for analysis.

In this paper, the values of each parameter are as follows. $\lambda = 1$, $\eta = 0.05$, υ=1/2, FSF is $\hat{p}(\omega) = e^{-0.5\omega^2}$, and the calculated κ is 23.5482.

4.2. FSWT in different network status

In this section, FSWT transformation is performed on the network traffic obtained in different network status, and the time–frequency distribution is obtained. Simultaneously, several representative characteristics are extracted from the time–frequency distribution to distinguish networks in different status better.

4.2.1. Different network status

We define Normal Status (NS) and the LDoS Attack Status (LS) for describing different network status. The TCP traffic gathered from a network bottleneck link free from LDoS attacks is represented by the network traffic in the NS. Furthermore, a sample of the network traffic in LS is provided by a bottleneck connection that is the target of LDoS attacks. Fig. 2 displays the network traffic gathered in the two network statuses.

As seen in Fig. 2, the distribution of network traffic in the NS is almost steady with very little oscillations. In contrast to the NS, the LS has a more distinct and highly fluctuating distribution of network traffic.

4.2.2. FSWT time–frequency distribution in different network status

First of all, for the convenience of analysis and discussion, the observation frequency is divided into two parts, namely the low-frequency part and the mid-high frequency part in this paper. The low-frequency part refers to the first 1/5 of the observation frequency, and the remaining 4/5 of the observation frequency is the mid-high frequency part. In this paper, the observation frequency is set to 50 HZ, then the low-frequency part is [0 HZ, 10 HZ], and the mid-high frequency part is [10 HZ, 50 HZ]. Then, we use FSWT to convert the network traffic into the time–frequency domain for further analysis. Fig. 3 displays the FSWT time–frequency distribution.

As Fig. 3 shows, in different network status, the FSWT time-frequency distribution covers the corresponding information of the time and frequency domains. The limitations of utilizing the time or frequency domain alone, which are introduced in Section 3, exist in practice. As depicted in Fig. 3(a) and (b), within the FSWT time–frequency distribution, the low-frequency part is significantly less than the mid-high frequency part under the NS. The FSWT time–frequency distribution in the LS is more concentrated in the low-frequency region, as shown by Fig. 3(c) and (d). The amplitude distribution is little in the low-frequency part because no LDoS attacks occur in the NS. However, it is concentrated in the mid-high frequency part with some noise. In the LS, due to the periodic LDoS attack in the network, the periodicity is concentrated more in the low-frequency area, which is in line with Chen et al. (2018) and Chen et al. (2005).

According to the above analysis, the FSWT time–frequency distribution under the NS has no obvious regularity in the low-frequency part. However, the low-frequency part under the LS is dominant in the overall FSWT time–frequency distribution. Among different frequency points, a relatively clear frequency band is presented when the LDoS attack occurs. Therefore, the low-frequency part of the FSWT time–frequency distribution can reflect different network statuses, and its performance characteristics are extracted to distinguish different network statuses in this paper.

4.2.3. The characteristics of FSWT time–frequency distribution in different network status

Based on the analysis above, the difference between the FSWT time–frequency distribution of network traffic under different network status is obvious. The entropy, the ratio of energy, contrast, and correlation are used to describe the difference in this paper. Let M_{ij} denotes the FSWT time–frequency distribution of network traffic, and the meaning and calculation formula of each characteristic are as follows.

The entropy (En) reflects the messy of the network FSWT time–frequency distribution. The FSWT time–frequency distribution is more complex and has a higher entropy value when it is more dispersed. The formula is as follows.

$$En = -\sum_i \sum_j M_{ij} \log M_{ij} \tag{3}$$

As is observed in Fig. 4 that the mean entropy of the FSWT time–frequency distribution in the NS is greater than the mean entropy in the LS. From Fig. 3(a) and (c), the FSWT time–frequency distribution

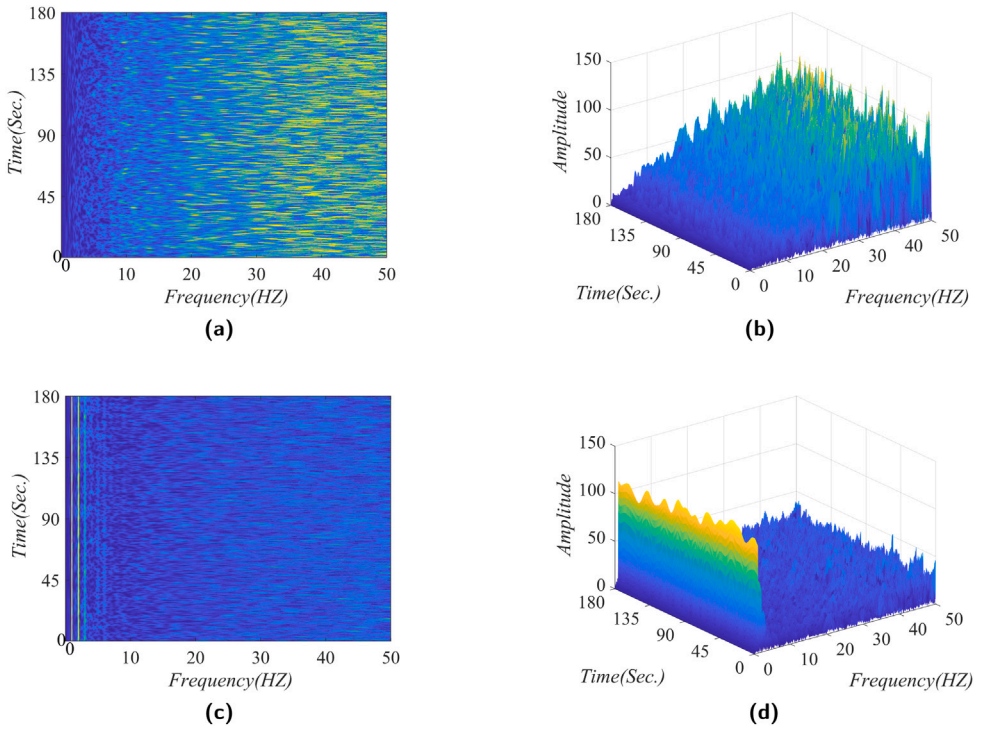


Fig. 3. The FSWT time–frequency distribution in different status. (a) is the 2D map of distribution in NS, and (b) is the 3D map of distribution in NS, (c) is the 2D map of distribution in LS, and (d) is the 3D map of distribution in LS.

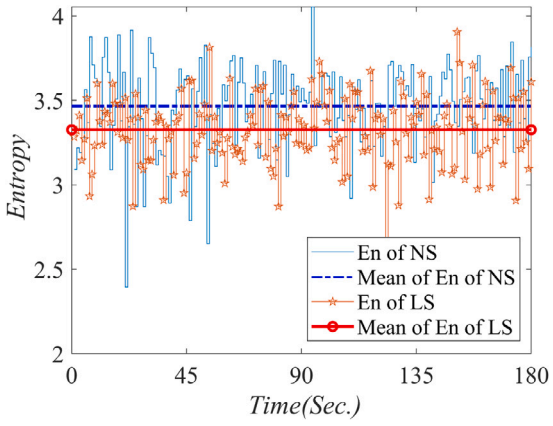


Fig. 4. The FSWT time–frequency distribution entropy in different status.

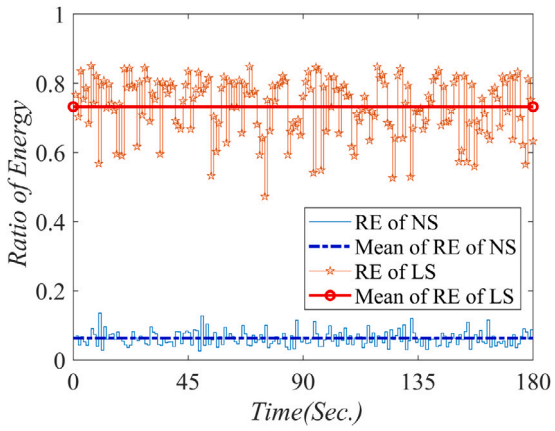


Fig. 5. Ratio of the FSWT time–frequency distribution energy in different status.

under the NS is relatively messy and complex, so its entropy value is larger than that in the LS. The FSWT time–frequency distribution in the LS is regular, so its entropy value is smaller than that of NS.

The ratio of energy (RE) is the ratio of energy between the low-frequency part and all frequency part. The larger the ratio is, the more the energy value is distributed in the low-frequency band. It is expressed by Eq. (4).

$$RE = \frac{\sum_i \sum_{\frac{1}{5}j} M_{ij}^2}{\sum_i \sum_j M_{ij}^2} \tag{4}$$

As depicted in Fig. 5, the ratio of energy in the FSWT time–frequency distribution in the NS is less than that in the LS. In particular, in the NS, the energy of the low frequency part is less than that in the LS, since the periodic LDoS attack is not contained in the NS, as shown in Fig. 3(a) and Fig. 3(c). However, the low-frequency portion of the LS has a higher energy ratio than the NS since the energy in the LS is mostly focused there.

Contrast (Con) reflects the degree of difference in the FSWT time–frequency distribution, representing the degree of its visual stripes. It is calculated by Eq. (5).

$$Con = \sum_i \sum_j (i-j)^2 M_{ij} \tag{5}$$

For the network traffic, the contrast value of its FSWT time-frequency distribution in different network statuses is shown in Fig. 6. The contrast of the FSWT time–frequency distribution under the NS is smaller than that under the LS. In Fig. 3(a) and (c), the FSWT time–frequency distribution under the NS is scattered, the distribution difference is not obvious, and the degree of visual strips is light, so the contrast is small. In the LS, the FSWT time–frequency distribution is comparatively concentrated, and the visual strips are dark, so the contrast is larger.

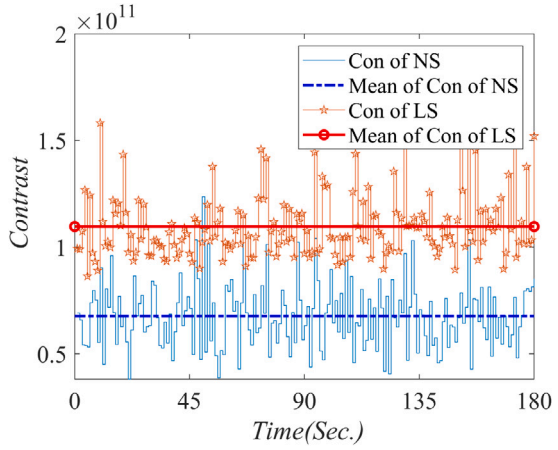


Fig. 6. FSWT time–frequency distribution contrast in different status.

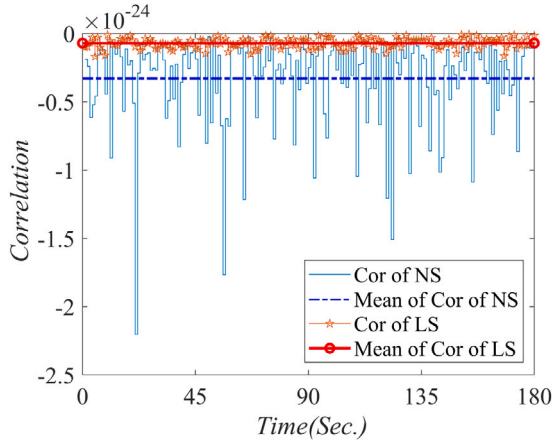


Fig. 7. The correlation of the FSWT time–frequency distribution in different status.

For the network traffic, the similarity of the FSWT time–frequency distribution among the row and column areas is reported by correlation (Cor) shown in Eq. (6), where $\mu_1 = \sum_i i \sum_j M_{ij}$, $\mu_2 = \sum_j j \sum_i M_{ij}$, $\sigma_1^2 = \sum_i (i-\mu_1)^2 \sum_j M_{ij}$, $\sigma_2^2 = \sum_j (j-\mu_2)^2 \sum_i M_{ij}$. The more stable the FSWT time–frequency distribution is, the greater the correlation value is.

$$Cor = \frac{\sum_i \sum_j ij M_{ij} - \mu_1 \mu_2}{\sigma_1^2 \sigma_2^2} \tag{6}$$

It can be concerned from Fig. 7 that the correlation of the FSWT the time–frequency distribution in the NS is smaller than that in the LS. From Fig. 3(a) and (c), the FSWT time–frequency distribution is messy and irregular under the NS, and the degree of similarity among the row and column areas is weak, so the correlation value is small. In the LS, the FSWT time–frequency distribution is relatively concentrated, and the similarity in the row and column areas is relatively greater, so the correlation is larger than that in the NS.

4.3. The LDoS attack detection based on FSWT time–frequency distribution

FSWT is used to get the time–frequency distribution. Effective characteristics are then retrieved in accordance with the time–frequency distribution of FSWT, and these characteristics are input into a trained classifier to produce detection results.

Table 1
Several common definitions.

	Detected: NS	Detected: LS
Actual: NS	True Negative (TN)	False Positive (FP)
Actual: LS	False Negative (FN)	True Positive (TP)

4.3.1. The detection framework

As shown in Fig. 8, the proposed LDoS attack detection framework has six main processes.

- The FSWT time–frequency distribution of the network traffic is obtained through the FSWT.
- According to the time–frequency distribution obtained by S1, four characteristics are extracted, including the entropy, the ratio of energy, contrast, and correlation.
- A decision tree model is trained using the features and labels of the training set.
- The training accuracy is verified to counter some problems. For example, overfitting may have occurred if the training accuracy is close to 100%, and some operations, e.g., pruning, need to be conducted for preventing overfitting.
- The characteristics of the testing data are extracted and input into the decision tree model trained in S4.
- The detection results of the testing data are obtained.

It is worth mentioning that the decision tree model is chosen as the classifier to detect the LDoS attack for the following reasons: (i) Fast inference speed. (ii) High accuracy. (iii) It can deal with continuous data with multiple categories. (iv) No field knowledge and parameter hypotheses are required. (v) Suitable for high-dimensional data.

4.3.2. Detection metrics

To better describe the detection effect of the proposed method, four detection metrics, accuracy, sensitivity, specificity, and precision are chosen for verification (David & Thomas, 2019).

Several common definitions are given in Table 1. For example, the True Negative (TN) indicates that the detection result is consistent with the actual status, i.e., the NS. Similarly, the definitions of FP, FN, TP are also shown in Table 1.

Accuracy ($Accu$) indicates the rate of correct detection in all data, and its formula is Eq. (7).

$$Accu = \frac{Num_{TP} + Num_{TN}}{Num_{TP} + Num_{FN} + Num_{FP} + Num_{TN}} \tag{7}$$

Sensitivity (TPR) computed as Eq. (8) means the probability that data is actually in the LS and correctly detected as the LS.

$$TPR = \frac{Num_{TP}}{Num_{TP} + Num_{FN}} \tag{8}$$

Specificity (TNR) is the probability that data is actually in the NS and correctly detected as in the NS.

$$TNR = \frac{Num_{TN}}{Num_{TN} + Num_{FP}} \tag{9}$$

Precision (Pre) represents the ratio of the data that is detected as in the LS and actually in the LS.

$$Pre = \frac{Num_{TP}}{Num_{TP} + Num_{FP}} \tag{10}$$

Besides, it is worth mentioning that the Detection Rate (DR) and Probability of Detection (P_D) mentioned in many papers is the TPR stated in this paper, the False Negative Rate (FNR) is related to the TPR, and the False Positive Rate (FPR) is related to the TNR in this paper. The relationship between them is as follows: $FNR = 1 - TPR$, $FPR = 1 - TNR$.

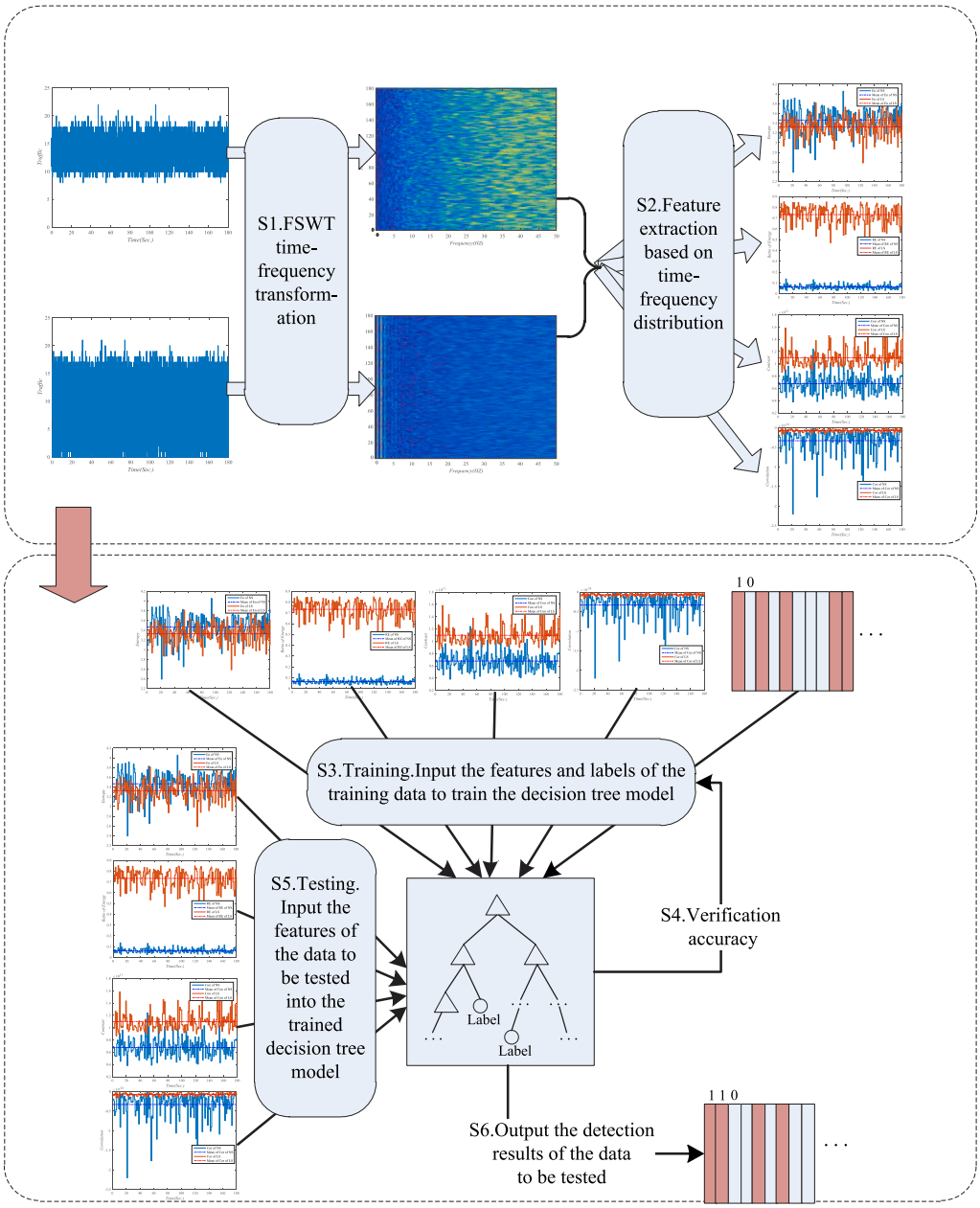


Fig. 8. The LDoS attack detection framework of this paper.

Among them, *FNR* represents the probability that the data is actually in the LS but is recognized as in the NS. *FPR* indicates the probability that the data is actually in the NS but is detected as in the LS.

5. Experiments

Firstly, as applying the time–frequency analysis on the network traffic, we verified the superiority of FSWT over STFT and WT. Then, we validated the effectiveness of our method with various dataset configurations using both the NS2 simulation platform and a physical testbed. To detect attacks in real time, we set up an SDN simulation platform and deployed several other methods for comparative experiments. Finally, our method also demonstrated acceptable detection performance on the Slowloris attack dataset.

5.1. Comparison of time–frequency analysis

We selected different window functions, window lengths, and movement steps to analyzing the network traffic in different environments. And we obtained the time–frequency distribution after STFT, as shown in Fig. 9. There are three traffic analysis groups in total. The parameter settings for each subgraph are listed in Table 2. As shown in Fig. 9(a), (b) and (c), STFT can indeed simultaneously capture the time–frequency information of network traffic under different environments, including the specific times and durations of frequency components. Fig. 9(c) and (d) use different window functions, and their time–frequency distributions show little difference at the start and end of the attack, but there is still a significant difference in the distribution of the low-frequency region. Fig. 9(d) and (e) have different window lengths, resulting in significant differences in their time–frequency distributions during the attack period, with the distribution in Fig. 9(e) being less distinct. Fig. 9(d) and (f) show considerable differences in time–frequency resolution due to different moving steps. Therefore,

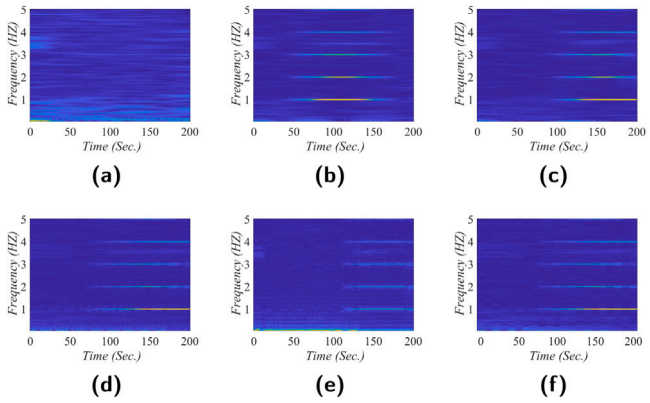


Fig. 9. STFT distribution of different traffic under various parameters.

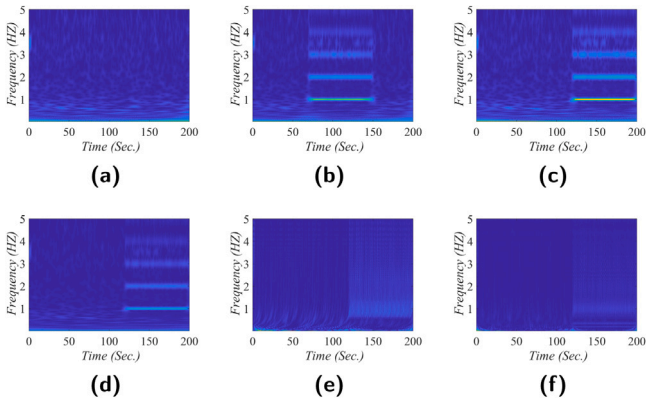


Fig. 10. WT distribution of different traffic under various parameters.

Table 2
Parameter settings for each subgraph in Fig. 9.

Subgraph	Traffic	Attack period	Window function	Window length	Moving step
a	1	No attack	Hamming	1024	10
b	2	70 s–150 s	Hamming	1024	10
c	3	120 s–200 s	Hamming	1024	10
d	3	120 s–200 s	Kaiser	1024	10
e	3	120 s–200 s	Kaiser	256	10
f	3	120 s–200 s	Kaiser	1024	100

Table 3
Parameter settings for each subgraph in Fig. 10.

Subgraph	Traffic	Attack period	Wavelet basis	Scale sequence length
a	1	No attack	cmor3-3	256
b	2	70 s–150 s	cmor3-3	256
c	3	120 s–200 s	cmor3-3	256
d	3	120 s–200 s	cmor3-3	1024
e	3	120 s–200 s	db12	256
f	3	120 s–200 s	bior3.3	256

the selection of parameters seriously affects the performance of STFT, limiting its application potential in distinguishing the network traffic.

Similarly, we selected different wavelet basis functions and scale sequence lengths to extract the time–frequency distribution of the network traffic under different environments through WT. There are three traffic analysis groups in total. The detailed settings are listed in Table 3. As shown in Fig. 10(a), (b), and (c), WT can obtain the time–frequency distribution of each group of network traffic. However, the effectiveness of the time–frequency distribution obtained by WT is still limited by the parameters of the wavelet basis and the scale sequence length, even if it is free from the constraints of fixed windows. In Fig. 10(c), (e), and (f), we used different wavelet basis functions, while in Fig. 10(c) and (d), we set different scale sequence lengths. And their time–frequency distributions exhibit significant differences, which can affect the results of the network traffic classification.

According to the settings of the frequency slice functions and energy coefficients in Table 4, we performed FSWT operation. The FSWT time–frequency distributions of the network traffic in different environments show significant differences. Notably, the normal and attack states are represented in Fig. 11(a) and (b) respectively. In network environments suffering from LDoS attacks, the FSWT time–frequency distributions exhibit good time–frequency resolution under different frequency slice functions and energy coefficient values. Additionally, there are clear and regular high-energy frequency bands during the attack periods, as shown in Fig. 11(b) to (f). This demonstrates that FSWT can maintain good time-domain and frequency-domain resolution despite varying parameter values, and it comprehensively describes the time–frequency distribution of the network traffic.

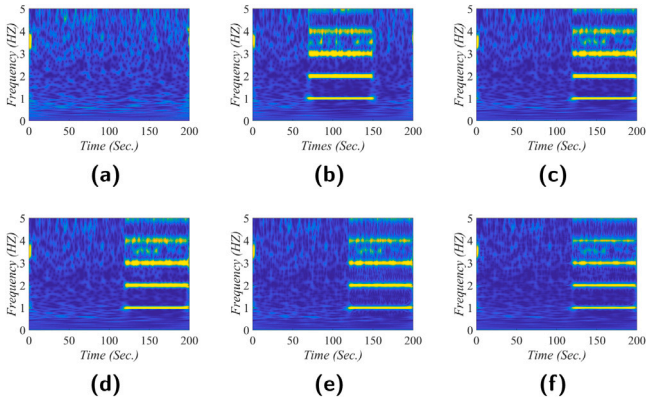


Fig. 11. FSWT distribution of different traffic under various parameters.

Table 4
Parameter settings for each subgraph in Fig. 11.

Subgraph	Traffic	Attack period	Frequency slicing function	Energy coefficient
a	1	No attack	$\hat{p}(\omega) = e^{-0.5\omega^2}$	1
b	2	70 s–150 s	$\hat{p}(\omega) = e^{-0.5\omega^2}$	1
c	3	120 s–200 s	$\hat{p}(\omega) = e^{-0.5\omega^2}$	1
d	3	120 s–200 s	$\hat{p}(\omega) = e^{-0.5\omega^2}$	2
e	3	120 s–200 s	$\hat{p}(\omega) = \frac{1}{1+\omega^n}$	1
f	3	120 s–200 s	$\hat{p}(\omega) = e^{-\lvert\omega\rvert}$	1

5.2. Experiment and analysis on NS2

A NS2 topology is built to obtain the network traffic, and our detection method is verified in this section.

5.2.1. Experimental environment and parameters

The network topology built on NS2 is shown in Fig. 12. The link between the two routers (R0 and R1) in the network is a 10 Mbps-bandwidth, 30 ms-delay bottleneck link. There are 10 legitimate users, which use TCP to communicate with the server all the time during the simulation, and the bandwidth of the rest links are 100 Mbps. An attacker periodically sends the UDP traffic to the target server at a certain rate during the attack interval. In addition, the link bandwidth and delay adjacent to the target server are also 100 Mbps and 15 ms respectively.

Based on the network in Fig. 12, we established three experimental sets, each with 9 subgroups. In Set 1, the attack period T is fixed to 1 s, while the attack duration L and attack intensity R vary across the subgroups. The range of L is [0.1, 0.2, 0.3] s, and the range of R is [20, 25, 30] Mbps. The simulation duration for each subgroup is 400 s, with

Table 5
Traffic dataset settings of experiments on NS2.

Set 1	LDoS attack parameters				Set 2	LDoS attack parameters				Set 3	LDoS attack parameters			
	T	L	R	Attack period		T	L	R	Attack period		T	L	R	Attack period
1	1 s	0.1 s	20 Mbps	140 s–260 s	1	2 s	0.1 s	20 Mbps	140 s–260 s	1	3 s	0.1 s	20 Mbps	140 s–260 s
2	1 s	0.1 s	25 Mbps	140 s–260 s	2	2 s	0.1 s	25 Mbps	140 s–260 s	2	3 s	0.1 s	25 Mbps	140 s–260 s
3	1 s	0.1 s	30 Mbps	140 s–260 s	3	2 s	0.1 s	30 Mbps	140 s–260 s	3	3 s	0.1 s	30 Mbps	140 s–260 s
4	1 s	0.2 s	20 Mbps	140 s–260 s	4	2 s	0.2 s	20 Mbps	140 s–260 s	4	3 s	0.2 s	20 Mbps	140 s–260 s
5	1 s	0.2 s	25 Mbps	140 s–260 s	5	2 s	0.2 s	25 Mbps	140 s–260 s	5	3 s	0.2 s	25 Mbps	140 s–260 s
6	1 s	0.2 s	30 Mbps	140 s–260 s	6	2 s	0.2 s	30 Mbps	140 s–260 s	6	3 s	0.2 s	30 Mbps	140 s–260 s
7	1 s	0.3 s	20 Mbps	140 s–260 s	7	2 s	0.3 s	20 Mbps	140 s–260 s	7	3 s	0.3 s	20 Mbps	140 s–260 s
8	1 s	0.3 s	25 Mbps	140 s–260 s	8	2 s	0.3 s	25 Mbps	140 s–260 s	8	3 s	0.3 s	25 Mbps	140 s–260 s
9	1 s	0.3 s	30 Mbps	140 s–260 s	9	2 s	0.3 s	30 Mbps	140 s–260 s	9	3 s	0.3 s	30 Mbps	140 s–260 s

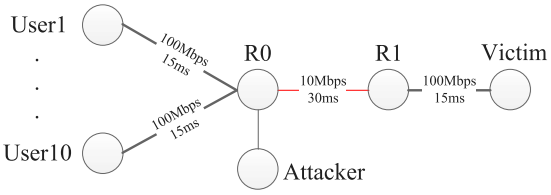


Fig. 12. The network topology based on NS2.

the LDoS attack occurring between 140 and 260 s, as shown in Table 5. Set 2 has a fixed attack period of 2 s, and Set 3 has a fixed period of 3 s. And the variations in L and R are the same as in Set 1.

We collected network traffic flowing through the bottleneck link at a sampling frequency of 100 Hz. The length of the detection window is set to 2 s. We extracted the FSWT time–frequency distribution of the window traffic to calculate relevant features to detect LDoS attacks. The experiment is divided into two parts, namely intra-set individual validation and inter-set cross validation.

5.2.2. Experiment and result analysis

Intra-set individual validation. We designated three subgroups from Set 1, Set 2, and Set 3 as training data, while the remaining six subgroups were divided into test data. This resulted in a total of 84 combinations of training and test splits for each experimental set. The values of the metrics Accu, TPR, TNR, and Pre for each set of experiments are shown in Fig. 13.

For Set 1, as shown in Fig. 13(a), the overall values of the four metrics across the 84 scenarios are generally stable and high, except for the result of subgroup 1, which is relatively poor. Subgroup 1 corresponds to the scenario where the experimental data subgroups 1, 2, and 3 are used as training data, and subgroups 4, 5, 6, 7, 8, and 9 are to be tested.

As shown in Fig. 13(b), the overall distribution of the four metrics across the 84 scenarios in Set 2 is stable, with the values consistently remaining at a high level.

The results of Set 3 is exhibited in Fig. 13(c). Although the evaluation metrics demonstrate stability, their overall performance is less optimal compared to Set 1 and 2. The primary difference is reflected in the values of TPR. Since the attack period for Set 3 is 3 s and the detection window length is 2 s, the attack period exceeds the detection window length, resulting in that some detection windows fail to identify the occurrence of LDoS attacks. Table 6 shows the results of the average Accu, average TPR, average TNR, and average Pre for each experimental set.

Inter-set cross validation. Three groups were chosen from each of Sets 1, 2, and 3 to create the training data, with the remaining experimental data serving as the test data. This allowed for the inclusion of different attack parameters in the training data. Thus, we obtained the detection results of inter-set cross validation shown in Table 7. The Accu values remain stable within the range [0.9642, 0.9683], with a range difference of only 0.0041. The TPR values are stable within the range [0.9296, 0.9398], with a range difference of 0.0102. The TNR values are stable within [0.9786, 0.9817], with a range difference of 0.0031. The Pre values are stable within the range [0.9490, 0.9565], with a range difference of 0.0075. These results indicate that our method demonstrates robust detection capability for LDoS attacks.

Table 6
The average metric values of three sets tested on NS2.

No.	Accu(avg)	TPR(avg)	TNR(avg)	Pre(avg)	Sum
Set 1	0.9915	0.9895	0.9923	0.9823	3.9556
Set 2	0.9855	0.9812	0.9874	0.9709	3.9250
Set 3	0.9392	0.8409	0.9813	0.9506	3.7120
Avg	**0.9721**	**0.9372**	**0.9870**	**0.9679**	**3.8642**

Table 7
Cross validation experiment results between sets.

No.	Train data	Accu	TPR	TNR	Pre	Sum
1	Set 1 (1,4,7) Set 2 (2,5,8) Set 3 (3,6,9)	0.9642	0.9306	0.9786	0.9490	3.8224
2	Set 1 (1,4,7) Set 2 (3,6,9) Set 3 (2,5,8)	0.9683	0.9398	0.9806	0.9539	3.8426
3	Set 1 (2,5,8) Set 2 (1,4,7) Set 3 (3,6,9)	0.9642	0.9296	0.9790	0.9499	3.8227
4	Set 1 (2,5,8) Set 2 (3,6,9) Set 3 (1,4,7)	0.9683	0.9370	0.9817	0.9565	3.8435
5	Set 1 (3,6,9) Set 2 (1,4,7) Set 3 (2,5,8)	0.9672	0.9380	0.9798	0.9521	3.8371
6	Set 1 (3,6,9) Set 2 (2,5,8) Set 3 (1,4,7)	0.9672	0.9361	0.9806	0.9538	3.8377
Avg		**0.9666**	**0.9352**	**0.9800**	**0.9525**	**3.8377**

5.3. Experiment and analysis on testbed

A network topology on the testbed environment is conducted to obtain the real network traffic. In this section, we validated the detection algorithm in an actual network environment.

5.3.1. Experimental environment and parameters

Real computers and routers are used to build a network topology to realize the real network environment. The testbed topology is shown as Fig. 15. The bottleneck link is designated as the connection between Routers 1 and 2. There are 3 legitimate users which are User1, User2, and User3. They send TCP connections to the target server all the time. During the LDoS attack interval, an attacker periodically transmits UDP attack packets to the target server at a specific rate. The bottleneck link is of 10 Mbps bandwidth, and the capacity of other links is 100 Mbps.

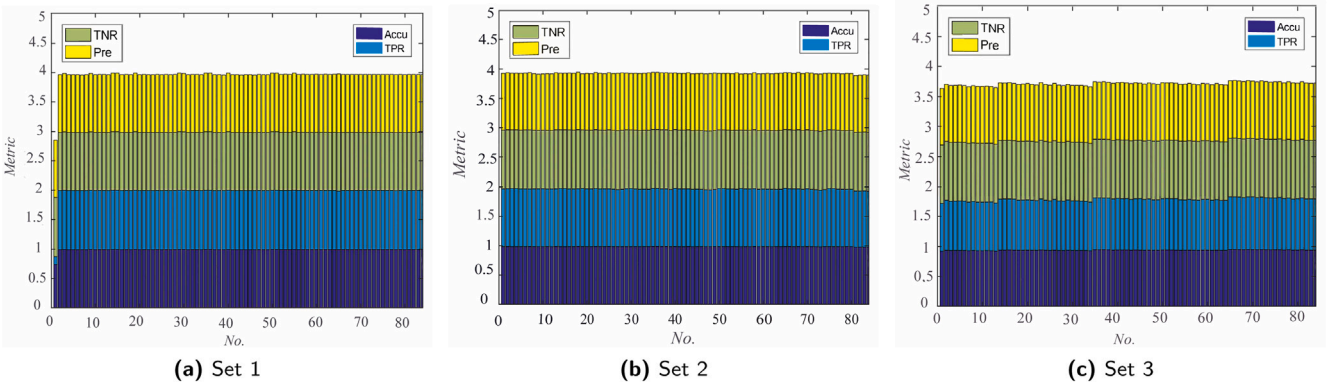


Fig. 13. The detection results of intra-set individual validation.

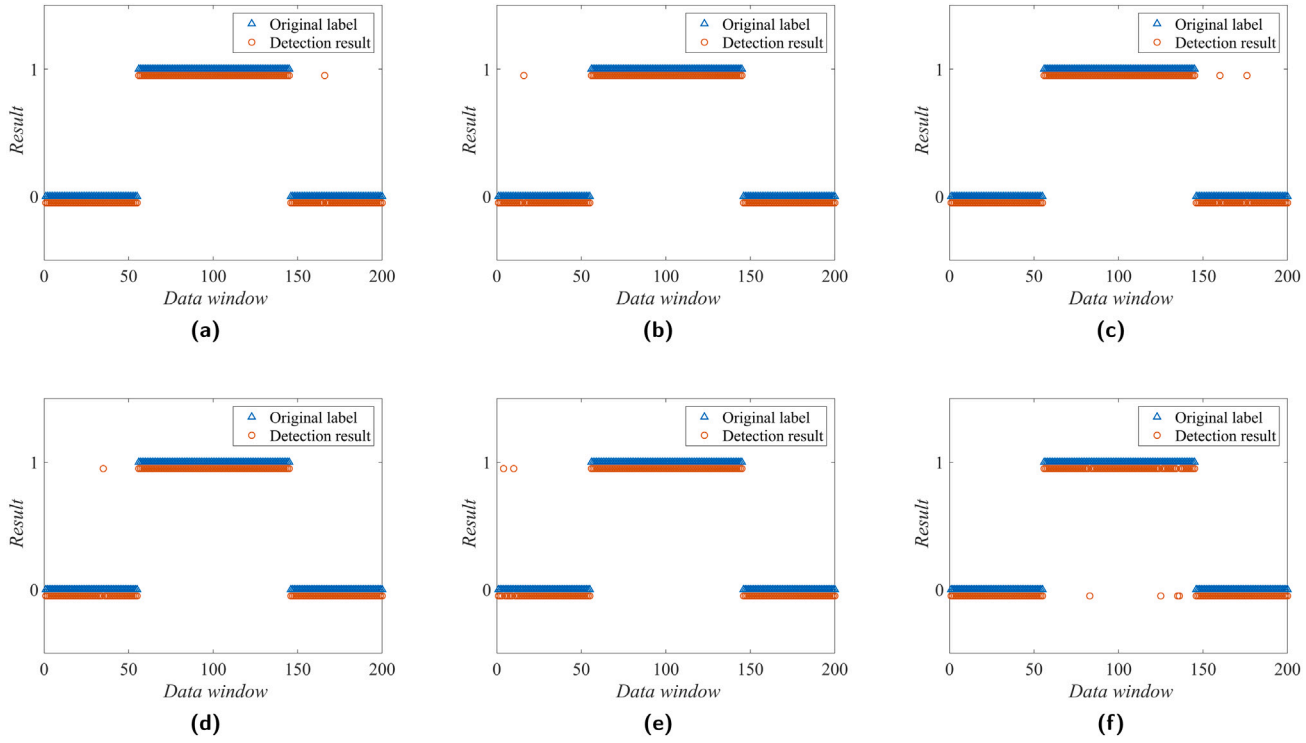


Fig. 14. The original labels and predicted results of experimental group numbers (2, 3, 5, 7, 9, 10) on testbed.

Table 8
The specific parameters of the 12 groups of experiments on testbed and SDN.

Group number	T (s)	L (s)	R (Mbps)
1	1	0.1	15
2	1	0.2	15
3	1	0.3	15
4	2	0.1	15
5	2	0.2	15
6	2	0.3	15
7	1	0.1	25
8	1	0.2	25
9	1	0.3	25
10	2	0.1	25
11	2	0.2	25
12	2	0.3	25

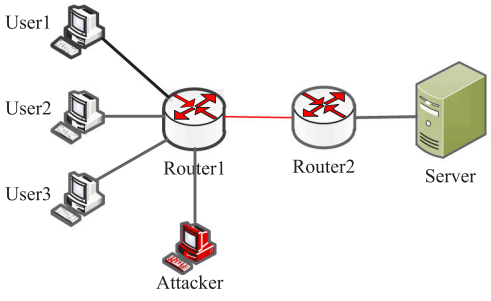


Fig. 15. The network topology based on testbed.

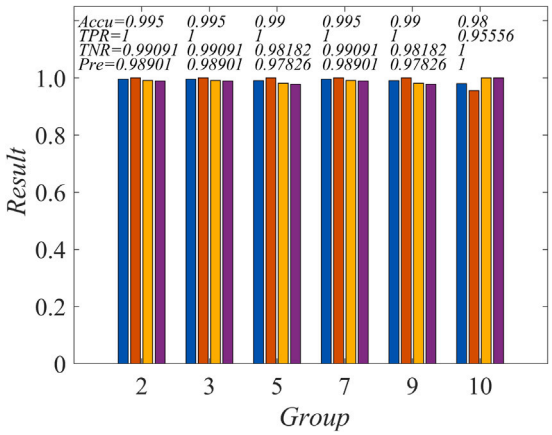


Fig. 16. The detection metrics of the 6 groups of testing data (2, 3, 5, 7, 9, 10) on testbed.

Based on the network topology of the testbed, 12 groups of experiments are designed, and the experimental time is 400 s for each group. The LDoS attacks occur during 110 s and 290 s. The LDoS attack parameters of each experiment are shown in Table 8. The network traffic is also sampled at a frequency of 100 HZ to obtain 12 groups of discrete traffic data for subsequent analysis.

5.3.2. Experiment and result analysis

Six groups of experiments (1, 4, 6, 8, 11, 12) are chosen to be trained as the training data. And the remaining six groups (2, 3, 5, 7, 9, 10) are utilized as the testing data to prove the effectiveness of our method.

The detection results of the experimental group numbers (2, 3, 5, 7, 9, 10) are shown in Fig. 14. In each subgraph of Fig. 14, '0' indicates the normal state NS, '1' represents the LDoS attack state LS, and the blue triangle shows the original labels of the testing data, and the orange circle exhibits the detection results. Based on the above results, the specific values of the detection metrics (accuracy, sensitivity, specificity, precision) of the 6 groups of testing data are shown in Fig. 16.

From Fig. 16, the accuracy values of the 6 groups are all greater than 0.98. The TPR value of group number 10 is 0.95556, and the TPR values of the other five groups are all 1. The TNR values of the 6 groups are all larger than 0.98. The precision value of the six groups ranges in (0.97,1]. This proves that our method is also effective in the testbed.

5.4. Experiment and analysis in SDN

We set up an SDN simulation platform and deployed several other methods for comparative experiments. Then the response time and cost of our method is provided.

5.4.1. Experimental environment and parameters

We built an SDN simulation environment by running the topology through Mininet (Mininet, 2010) and establishing connections with the Ryu (Ryu, 2010) controller. The Napnet topology provided by topology zoo (Knight, Nguyen, Falkner, Bowden, & Roughan, 2011) dataset is a real-world topology among six cities in the United States. Every city node in Fig. 17 indicates a switch and a host that is linked to it. Except for the link between San Jose and Vienna, which has a bandwidth of 45 Mbps, all other links have a bandwidth of 128 Mbps. We set the host located in San Jose as the victim and launched an LDoS attack from Vienna to congest the link between them.

To closely simulate a real-world scenario, we replayed the UDP traffic from the WIDE (MAWI, 2023) dataset as background UDP traffic. And TCP connections started by Iperf (Iperf, 2023) occupy the remaining bandwidth. The attack host launched the LDoS attacks using

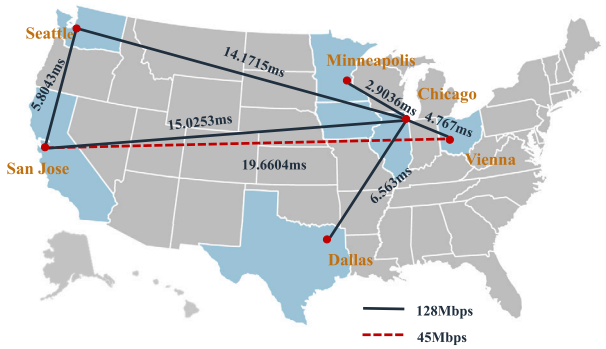


Fig. 17. Napnet topology in topology zoo.

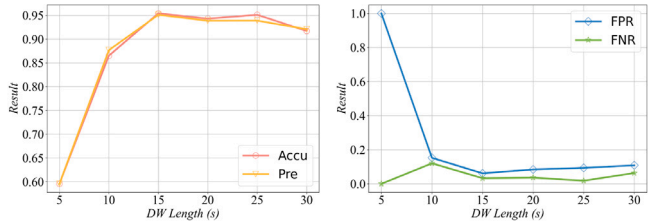


Fig. 18. The influence of DW length on detection results.

Table 9
Comparison with other detection methods.

Method	Accu	Pre	TPR	TNR
ERT-EDR (Liu et al., 2024)	0.9316	0.8779	1.0000	0.8655
GASF-IPP (Tang, Wang, Liu et al., 2023)	0.9134	0.9268	0.8956	0.9309
AI-based IDS (Chang et al., 2023)	0.9184	0.9052	0.9326	0.9046
Our method	0.9541	0.9511	0.9669	0.9338

a Python script. The attack parameters and the division of training and test data are consistent with the settings in Section 5.3. The detection method is deployed on the Ryu controller, which polls switches to collect traffic information and calculate features in units of window. The trained model is loaded to determine the window state, enabling real-time detection of LDoS attacks.

5.4.2. Experiment and result analysis

Detection Window settings. The controller requests flow information from the switches every 0.5 s and establishes a detection window (DW) with a sliding step of 2 s. The experimental results for different window lengths are shown in Fig. 18. To better present the results, the trends of FNR and FPR are displayed, which are opposite to the trends of TPR and TNR. When the window length is very small, the traffic sequence within the detection window is too brief, rendering time–frequency analysis ineffective and resulting in meaningless classification results. An excessively long window length causes the traffic information inside it to not refresh rapidly enough, which raises the FPR. For the subsequent experiments in this paper, a 15 s DW is used by default.

Method comparison. ERT-EDR (Liu et al., 2024) and GASF-IPP (Tang, Wang, Liu et al., 2023) designed detection methods for LDoS attacks in SDN based on timing analysis. An AI-based IDS (Chang et al., 2023) proposed an intrusion detection framework that combines various machine learning methods and used principal component analysis to select important features. Under the same environment and dataset settings, we implemented these methods. Table 9 shows the results, which indicates that our method outperforms the others.

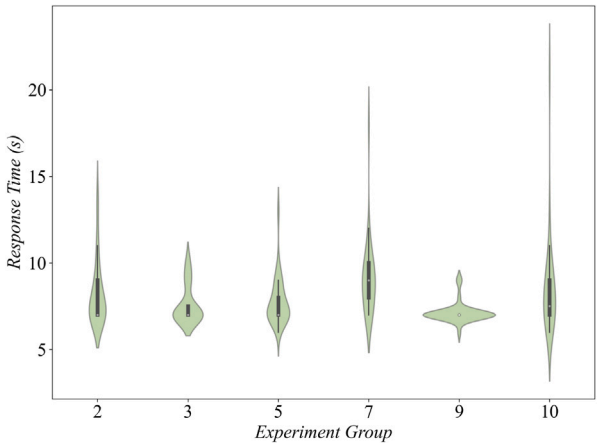


Fig. 19. Response time.

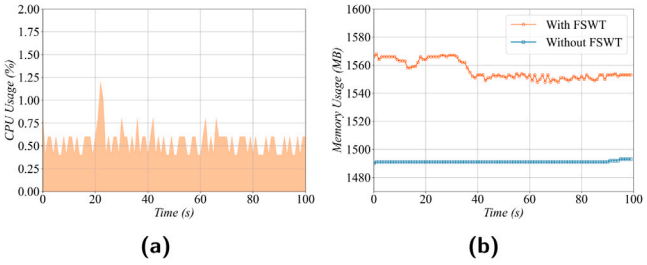


Fig. 20. The CPU and memory overhead of this method when running on the Ryu controller.

Response time. In order to reduce incorrect alarms, the controller will only decide that an LDoS attack is occurring if three consecutive DWs are identified as attack windows. The response time is calculated as the difference between the initiation of the attack and the controller's determination that the attack occurred. The results of 30 repetitions of each group of attack experiments are displayed in Fig. 19. Among these 180 experiments, the fastest attack determination took 6 s, the slowest took 21 s, and the average response time was 8.06 s.

Resource usage. We measured the overhead of our method when deployed on the Ryu controller, as shown in Fig. 20. The CPU usage rate remains below 1.2%, and the memory overhead remains stable at around 65 MB. This indicates that our method can be deployed as a lightweight application in practice.

5.5. Detection of Slowloris Attacks

A comparable kind of slow-rate attack is the Slowloris (Rios, Inacio, Magoni, & Freire, 2024) attack. At a very low pace, the attacker sends valid but incomplete HTTP requests to the target site. To maintain connectivity, these requests are sent periodically and intermittently, consuming the number of concurrent connections that the server can support. The CIC-IDS2017 (Sharafaldin, Lashkari, Ghorbani, et al., 2018) dataset includes a 23-minute Slowloris attack. We extracted the packet PCAP for this time period and divided it into 2-second slices, resulting in a total of 690 slices. By applying the method proposed in this paper, we identified 579 attack slices among them. This indicates that our method is scalable and useful for identifying similar slow-rate attacks.

6. Discussion

6.1. Network security application

Computer network security needs to ensure the confidentiality, integrity, and availability of relevant resources. As a form of boundary protection, firewalls can block suspicious traffic at the network edge. Our method can work at the network layer as a packet filter firewall to detect malicious occupation of bandwidth resources. Intrusion Detection and Prevention Systems (IDPS) can be directly deployed behind the firewall to scan incoming traffic against security threats, promptly alerting and taking protective measures (Chang et al., 2023). Even if the aforementioned two-level defense in depth is adopted, there is still a risk of being cleverly bypassed by hackers. Therefore, enhancing the security of the host itself is crucial, which can be achieved through application monitoring systems and operating system security enhancements. High-performance attack detection can be implemented on the host by combining it with Data Plane Development Kit. Additionally, in a zero-trust framework that emphasizes internal network security, authorization and access control mechanisms can be introduced to establish a zero-trust network.

6.2. Network architecture

Specific network architectures can address the performance and management requirements of different network scenarios. Software-Defined Networking (SDN) separates the control plane from network devices, enabling flexible and centralized network management and operation. In large data centers, controllers can uniformly schedule the flow of business traffic based on the information from network elements, ensuring both network reliability and scalability. Moreover, SDN facilitates the deployment and dynamic adjustment of security policies and applications, allowing for the automatic isolation or redirection of threat traffic. As the control plane becomes critical to the network, the requirements for performance and interaction link reliability increase (Tang, Yan, Gao, Liang and Jin, 2023). Programmable forwarding devices with line-rate forwarding capabilities allow operators to customize packet parsing and forwarding behaviors, thereby carrying some security policies and mechanisms to reduce the pressure on the control plane. However, the limited storage resources and instruction sets of the domain-specific chips in these programmable devices increase the complexity to the design and deployment of in-band security policies. Currently, for traffic classification and filtering, the industry is leaning towards intelligent data planes powered by the machine learning algorithms (Swamy, Rucker, Shahbaz, Gaur, & Olukotun, 2022; Zheng et al., 2022).

The Internet of Things (IoT) is a network composed of sensor nodes, applicable in systems such as intelligent transportation and smart home systems. Due to the limited computational, storage, and communication capabilities of these nodes, they are particularly vulnerable to threats such as LDoS attacks. Such attacks can disrupt the normal operations of IoT devices, leading to significantly severe security threats and operational challenges, especially in critical infrastructure environments. Our method features a simple detection process with low computational and storage complexity, making it an ideal solution for protecting IoT systems from LDoS attacks (Wu, Cheng, Hsu and Wu, 2019), which helps to improve the overall resilience and reliability of IoT based systems.

Notably, the rapid development of mobile IoT intensifies the burden on multi-access edge computing (MEC) nodes, which transmit data with a vast number of mobile devices. Once the MEC nodes crash, it will cause security issues such as the leakage of private data. Blockchain-based authentication technology can be used to ensure secure communication between nodes. However, it faces challenges such as limited on-chain resources, exposure of node trajectories, and frequent data uploads. BUA (Wen, Liu, Yang, Yang, & Chang, 2024) addresses these

issues by designing an Unlinkable Authentication Protocol and utilizing vector commitment compression technology to achieve the three key goals of anonymity, unlinkability, and mutual authentication. It is an excellent authentication mechanism for mobile IoT.

6.3. Limitation

We only conducted research and analysis on single-source LDoS attacks and developed a detection method accordingly. However, distributed LDoS attackers utilize multi-source coordinated attack strategies, ultimately aggregating attack streams at the victim target. This attack method has higher attack intensity and more complex patterns increasing the difficulty of tracing the source. Currently, there is a lack of a comprehensive detection system for both synchronous and asynchronous distributed LDoS attacks. Although our method can be regarded as a universal LDoS attack detection mechanism that can be widely deployed on multiple platforms, it lacks specific guidance on how to achieve optimal deployment and maximize effectiveness for a specific network environment, such as wireless sensor networks, cloud computing platforms and programmable data planes. In addition, it is necessary to establish an integrated framework for detection and defense, which can filter attack flows by flow identifiers such as IP addresses to reduce damage.

7. Summary and future work

Aiming at the periodical and stealthy LDoS attack, the existing LDoS research processes and technologies are summarized. The limitations of the mainstream detection methods are discussed in this paper. The fact is found that more detailed information is contained in the time–frequency domain, so we extracted the characteristics of the LDoS attack information in the time–frequency domain to make the detection more accurate. However, there are some limitations of the existing LDoS research based on known time–frequency domain, such as STFT and WT. To this end, a more stable time–frequency domain transformation FSWT is used to analyze the network traffic of different network status. Four relevant characteristics (the entropy, ratio of energy, contrast, and correlation) are extracted, and the decision tree is chosen to be trained for detecting the testing data.

We conducted experiments using different topologies and datasets in NS2, testbed, and SDN. The detection accuracy of this method can reach over 95% in all three environments, and the response speed is fast. It can be deployed in a lightweight manner, and its detection ability against slowloris attacks indicates its scalability potential. In the future, the behavior of LDoS attacks and other types of attacks in the time–frequency distribution will be explored to better characterize LDoS attack information. Only in this way can the extracted LDoS attack characteristics be more accurate, and the detection accuracy can be further improved.

CRediT authorship contribution statement

Xiaocai Wang: Conceptualization, Methodology, Software, Validation, Visualization, Formal analysis, Investigation, Writing – review & editing , Writing – original draft. **Dan Tang:** Resources, Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing. **Ye Feng:** Conceptualization, Methodology, Software, Visualization, Validation, Formal analysis, Investigation, Writing – original draft. **Zheng Qin:** Supervision, Conceptualization, Writing – review & editing. **Bing Xiong:** Supervision, Investigation, Writing – review & editing. **Yufeng Liu:** Supervision, Writing – review & editing.

Declaration of competing interest

All authors disclosed no relevant relationships.

Data availability

No data was used for the research described in the article.

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